

pre_place.tcl

Description : Pre-place script – boundary/end-cap/tap/IO/spare-cell setup.

Tool : Synopsys ICC2 (or compatible implementation tool)

Requirements : tech_setup.tcl, user_setup.tcl, floorplan stage completed

Variables :

max_core - Max CPU cores

ndm_lib_name - Library name under db/

design - Top-level design name

end_cap_cell_l/r/b/t- End-cap cell names (left/right/bottom/top)

tap_cell - Well tap / substrate tap cell name

tap_cell_distance - Tap insertion distance

io_buf - IO/boundary buffer cell name

stop_after_pre_place- Boolean; if 1, stop after pre-place

Read Technology & User Settings

source -e -v ./scripts/tech_setup.tcl

source -e -v ./scripts/user_setup.tcl

Multi-CPU Setting

set_host_options -max_cores \$max_core

Open NDM/NDM Library and Create Pre-Place Block

if { [file exists "db/\${ndm_lib_name}"] } {

puts "\n[INFO] Library 'db/\${ndm_lib_name}' exists, opening...\n"

open_lib db/\${ndm_lib_name}

copy_block -from \${design}_floorplan_done -to

\${design}_pre_place_done

current_block \${design}_pre_place_done

link

} else {

puts "\n[ERROR] NDM Library 'db/\${ndm_lib_name}' does not exist.

Please check.\n"

return

}

End-cap Cells

```
puts "\n[INFO] Inserting boundary/end-cap cells...\n"
```

```
set end_cap_cell_left [get_lib_cells */$end_cap_cell_l]  
set end_cap_cell_right [get_lib_cells */$end_cap_cell_r]  
set end_cap_cell_bottom [get_lib_cells */$end_cap_cell_b]  
set end_cap_cell_top [get_lib_cells */$end_cap_cell_t]
```

```
set_boundary_cell_rules \  
-left_boundary_cell $end_cap_cell_left \  
-right_boundary_cell $end_cap_cell_right \  
-bottom_boundary_cell $end_cap_cell_bottom \  
-top_boundary_cell $end_cap_cell_top \  
-prefix END_CAP
```

```
compile_boundary_cells
```

```
# Fix boundary cells so they are not moved  
set_fixed_objects [get_cells -hierarchical *boundary*]
```

```
# Tap Cells  
puts "\n[INFO] Inserting tap cells...\n"
```

```
set tap_cell_name [get_lib_cells */$tap_cell]
```

```
create_tap_cells \  
-lib_cell $tap_cell_name \  
-distance $tap_cell_distance \  
-pattern stagger \  
-skip_fixed_cells
```

```
set_fixed_objects [get_cells -hierarchical *tap*]
```

```
# IO Buffers / Boundary Buffers  
puts "\n[INFO] Inserting IO/boundary buffers on all signal ports...\n"
```

```
set io_buf_name [get_lib_cells */$io_buf]
```

```
foreach_in_collection each_port [get_ports * -filter "port_type == signal"] {
    set port_name [get_object_name $each_port]
    insert_buffer -new_cell_names io_buf_${port_name} \
        [get_ports $port_name] $io_buf_name
}
```

Apply Hard Keepout to IO Buffers

puts "\n[INFO] Applying hard keepout margin around IO buffers...\n"

```
create_keepout_margin \
    -outer {0.152 0 0.152 0} \
    -type hard \
    [get_cells -hierarchical *io_buf*]
```

Pull IO Buffers Close to IO Ports

puts "\n[INFO] Pulling IO buffers closer to IO ports (magnet placement)...\n"

```
magnet_placement \
    -logical_level 1 \
    [get_ports *] \
    -mark_fixed \
    -move_legalize_only
```

Mark IO Buffers as Fixed and dont_touch

```
set_fixed_objects [get_cells *io_buf*]
set_dont_touch [get_cells *io_buf*] true
```

Spare Cells (Example: ~2% Addition)

Customize the list and counts per your library/requirements.

puts "\n[INFO] Adding spare cells...\n"

```
add_spare_cells \
    -num_cells {DFFX1_RVT 1 NAND2X4_RVT 2 NOR2X4_RVT 2 INVX4_RVT 5}
\
    -cell_name SpareCell \
    -input_pin_connect_type tie_low \
```

```
-repetitive_window {80 80} \  
-ignore_blockage_types {soft partial}
```

```
set_attribute [get_cells *Spare*] is_spare_cell true  
set_dont_touch [get_cells *Spare*] true
```

```
# Save the Block
```

```
puts "\n[INFO] Saving block after pre-place operations...\n"  
save_block
```

```
# Exit or Stop After Pre-Place
```

```
sh touch pre_place_done
```

```
if { $stop_after_pre_place } {  
    puts "\n[INFO] Pre-place stage completed (stop_after_pre_place = 1).  
Returning...\n"  
    return  
} else {  
    puts "\n[INFO] Pre-place stage completed. Closing design and exiting  
tool...\n"  
    close_blocks -force  
    close_lib  
    exit  
}
```